Session Objectives

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- Define and recognize the symptoms of asthma
- Understand the epidemiology of asthma and identify the different types of asthma
- Describe the relationship of IgE to asthma and distinguish the various roles that environment and genetics play in asthma
- Detect mimickers of asthma
- Recognize status asthmatics and warning signs for severe asthmatics
- Explain the clinical presentation and tests necessary to diagnose asthma and come up with a differential diagnosis
- Know the goal of asthma treatment and management

Indicates tested on the boards



"As we have become more and more dependent upon the laboratory and special investigations in making our diagnosis, we have gradually lost that faculty of clinical observation which enabled these early clinicians to make an accurate diagnosis in the more common diseases almost with a glance at the patient."

Dr. Will Mayo

History of the Diagnosis of Asthma

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- In 1907 all agreed that a neurosis is an important factor, but many believe that is also an inflammatory affection of the bronchioles that is either the primary or secondary cause of this condition
  - Clinically the disease presents in three fundamental symptoms
    1. The paroxysmal charter of the attacks
    2. The dyspnea
    3. The vasosecretory disturbance

Current Definition of Asthma

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- "Asthma is a **heterogeneous** disease, usually characterized by chronic airway inflammation. It is defined by the history of the respiratory symptoms such as wheeze, shortness of breath, chest tightness and cough that vary over time and in intensity, together with variable expiatory airflow limitation" - Global Initiative for Asthma 2017
- An obstructive lung disease that is ***reversible***

Defining Asthma

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- Symptoms that **increased** the probability for asthma diagnosis
  - More than one of the classic symptoms (wheeze, SOB, cough, chest tightness)
  - Symptoms worse and night and early morning
  - Symptoms vary over time and vary in intensity
  - Symptoms are triggered by viruses, exercise, allergens, changes in weather, laughter or irritants (smoke, strong smells, fumes)
- Symptoms that **decrease** the probability for asthma diagnosis
  - Isolated cough with no other respiratory symptoms
  - Chronic sputum production
  - SOB associated with dizziness, light-headedness or peripheral tingling
  - Chest pain
  - Exercise-induced dyspnea with noisy inspiration



Current Classifications of Asthma

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- **Allergic (Atopic) asthma**
  - Personal and/or family history of allergic diseases
  - Atopy
    - Positive skin tests
  - Increased levels of serum IgE
  - Positive response to bronchoprovocation with specific antigen

- **Idiosyncratic asthma**
  - No personal and/or family history of allergic diseases
  - Non-atopic
    - **Lack of positive skin tests to aeroallergens**
  - Normal serum levels of IgE
  - Examples: exercise induced asthma, aspirin intolerant asthma, obesity, and occupational exposure

Older categories: intrinsic asthma and extrinsic



Phenotypes of Idiosyncratic Asthma

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- **Exercise induced asthma**: caused by loss of heat, water, or both from lungs during exercise to the cooler or drier environmental air
  - Occurs within 10 minutes of starting, peaks after 20-30 minutes
  - May be a marker for poorly controlled asthma
  - In 11% of Olympic athletes
  - Diagnosis: exercise challenge test: ↑ HR to 80% of max for 4-6 mins
    - **Positive test**: ↓ in FEV1 (or PEF) after exercise
      - Measurements taken at 5 minute intervals for 20-30 minutes post-exercise
  - Treatment goal: participation in ANY physical activity without limitation
    - Longer warm-up periods are shown to be preventative
    - B2- agonists prevent >80% of exacerbations when used BEFORE activity



Phenotypes of Idiosyncratic Asthma

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- **Aspirin Intolerant Asthma (AIA) or Aspirin Exacerbated Respiratory Disease (AERD):** typically presents in adults within a three hour window of taking aspirin
  - Triad: asthma, nasal polyps and aspirin use
  - Symptoms 20 minutes to 3 hours after taking ASA
  - Estimated 5.5-12% of adults with asthma
- **Asthma with obesity**: elevated body mass index (BMI) and little eosinophilic airway inflammation
- **Occupational Exposure**: 5-10% of adult onset asthma
  - Variety of different exposures discovered; key to diagnosis is positive methacholine challenge test after exposure to specific agent



Etiology of Asthma

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A chronic heterogeneous disease of the airways. An individual with asthma will **always have** asthma but *will not always demonstrate* symptoms of an asthma.

Inflamma-tion of the lungs

Type 1 Hypersensitivity Reaction

Hyper-respon-siveness to stimuli

Airflow obstruction

reversible

Signs and symptoms of Asthma

Asthma exacerbation

Wheezing
Breathlessness
Chest tightness
Nighttime or early morning cough

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Pathophysiology of Asthma: Airflow Obstruction

- The *smooth muscle dysfunction* with the increased **edema** and **mucous** lead to **obstruction** of the airway and symptomatic asthma:
  - Wheezing
  - Chest tightness
  - Cough
  - Shortness of breath
  - Trapping of CO2



Phases of Asthma

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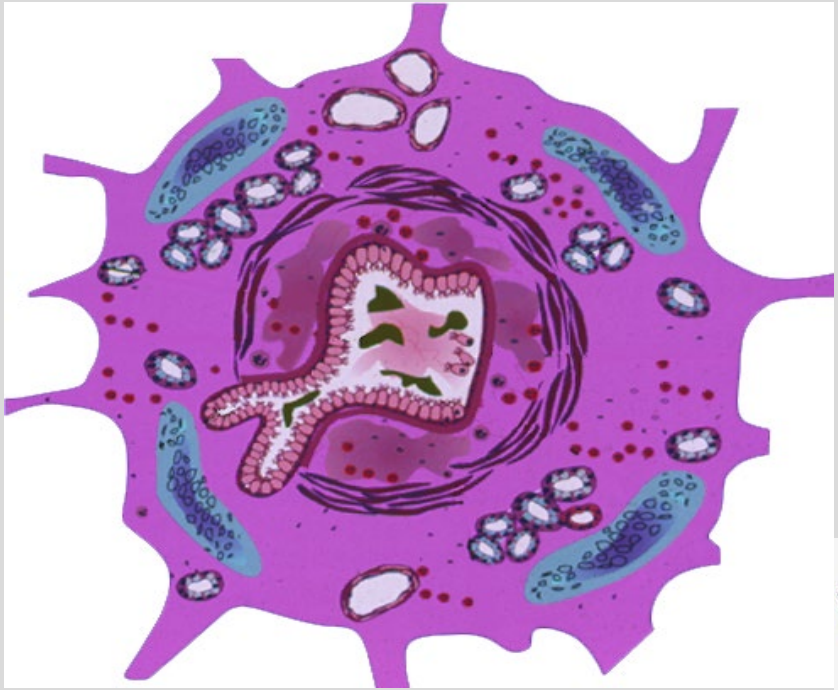
- Acute Response
  - Bronchial hyperactivity
  - Mucosal edema
  - Airway secretions
- Chronic Inflammatory Response
  - Increased inflammatory cell numbers
  - Epithelial damage
  - Airway remodeling



Chronic Changes of Asthma: Airway Remodeling

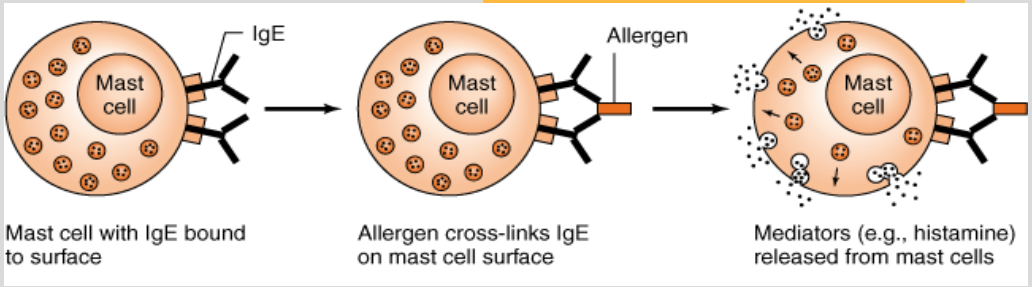
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- Basement membrane thickening
- Vascular Dilation
- Edema (due to ↑ permeability)
- Inflammatory Cell Infiltration
  - Eosinophils, T-lymphocytes, neutrophils
- Epithelial Damage
- Smooth Muscle Hypertrophy
- Mucous gland hypertrophy and hyper secretion
- *Parenchyma is spared*





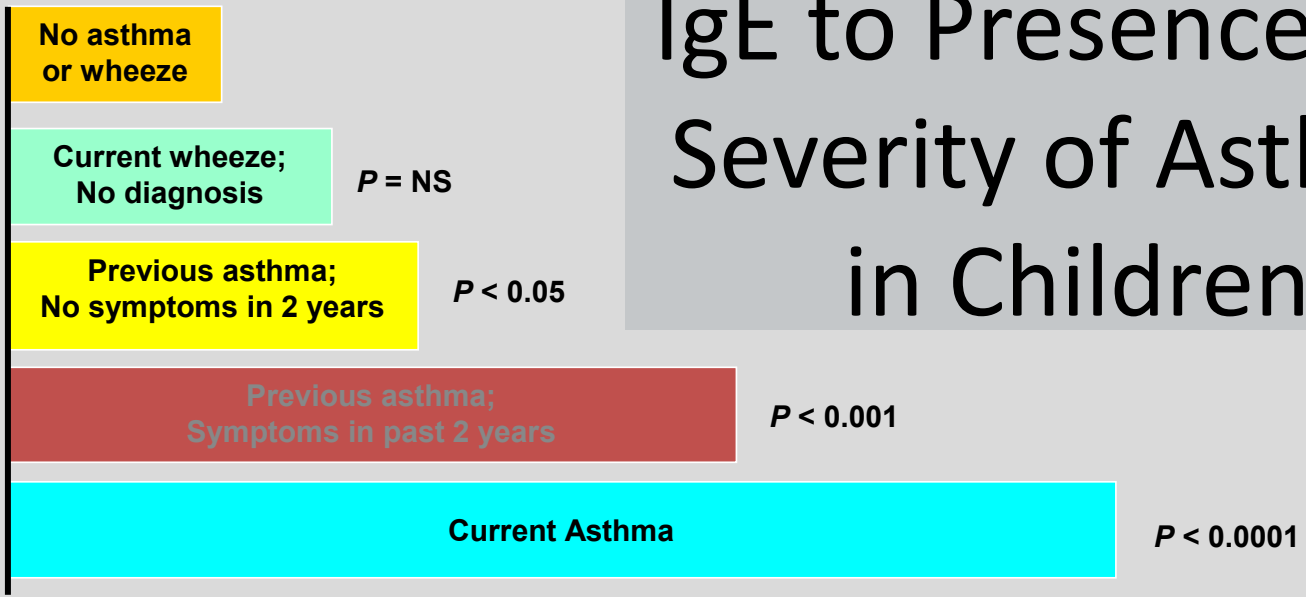
IgE and Asthma



- "Atopy, or the genetic predisposition to develop specific IgE antibodies directed against common environmental allergens, is the strongest identifiable risk factor for the development of asthma"
- IgE binds to high-affinity (FcεRI) receptors on mast cells and basophils and triggers the release of inflammatory mediators
- Increased IgE is associated with increased asthma prevalence and increased airway hyperresonspiveness
  - Low levels of IgE due not necessarily indicated lack of allergy
- Other causes for ↑ eosinophils NAACP (Neoplasia, Asthma, Allergy, Chronic adrenal insufficiency, Parasites [*invasive*])



Relation of Serum IgE to Presence and Severity of Asthma in Children

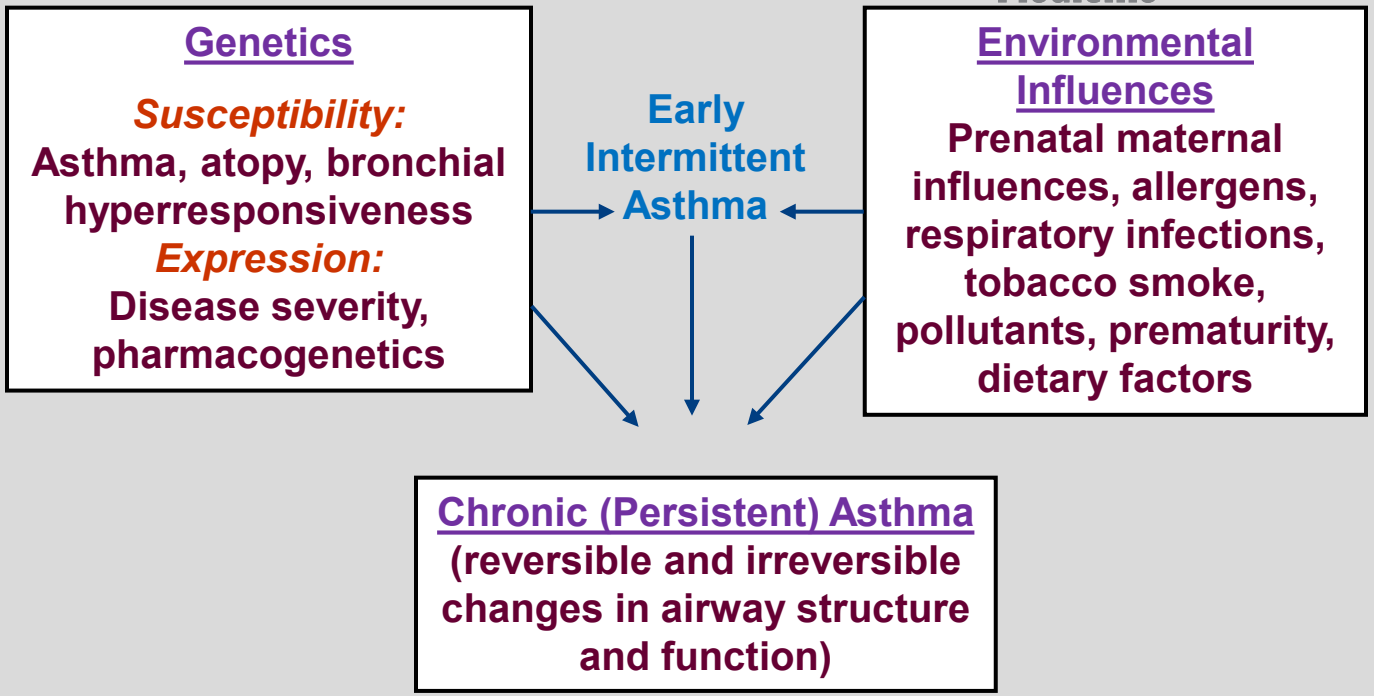


P values reflect comparison to group with no asthma or wheeze.

Sears MR et al. *N Engl J Med.* 1991;325:1067-71.

Gene-environment Interactions in Asthma Development

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Courtesy of E.R. Bleeker Jr., MD

Diagnosing Asthma

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- No precise diagnostic test exists
  - "All that wheezes isn't asthma"
  - "Not all asthmatics wheeze"
- **To make the diagnosis you need**:
  1. Detailed medical history and physical that is suggestive of asthma
  2. Demonstration of variable expiratory airflow limitation
  3. Exclusion of alternative diagnoses

Diagnosing Asthma: History and Physical

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- **Symptoms of coughing, wheezing, SOB, chest tightness**
  - Coughing at night or after physical activity
  - Colds that last over 10 days
- **Pattern of symptoms**
  - Time of day? Time of year? With an exposure?
- **Severity of symptoms**
- Diagnosed with atopic dermatitis, eczema or other allergic skin conditions
- **Family history**
- **Relief when medication is used**

Diagnosing Asthma: History and Physical

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- **Physical Examination**
  - Wheezing
  - Tachycardia
  - Mild hypertension (systolic)
  - Nasal findings: polyps, edema, discharge,
  - Skin: eczema, atopic dermatitis
  - **Signs of SEVERE Airflow Obstruction**
    - Paradoxical pulse
      - ↓ in amplitude of systolic BP by >10 mmHg during inspiration; seen in cardiac tamponade, asthma, obstructive sleep apnea, pericarditis and croup
    - Hunched appearance
    - Use of accessory muscles
    - **Silent chest!**



Diagnosing Asthma: Testing Expiratory Airflow

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- **Spirometry**
  - A test used to measure the **VARIABILITY** lung function and to determine the expiratory airflow capacity
    - *Since asthma is a reversible airway obstruction it is key that this value changes during an acute asthmatic exacerbation and returns to baseline post-treatment*
  - FEV_1 (% of predicted normal)
    - At risk or mild ≥80%
    - Mildly reduced: 60-79%
    - Moderately reduced: 40-59%
    - Severely reduced: 30-49%
  - **This test by itself is NOT adequate for diagnosis**

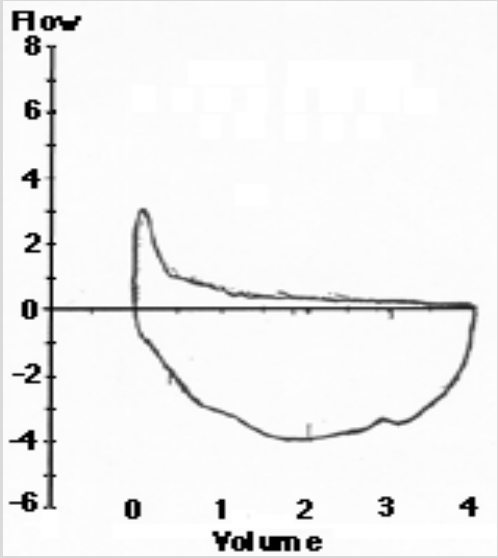


Diagnosing Asthma: Testing Expiratory Airflow

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- **Obstructive Disease Flow Volume Loop**
  - ***The SCOOP***



↓↓ FEV_1, ↓ FVC

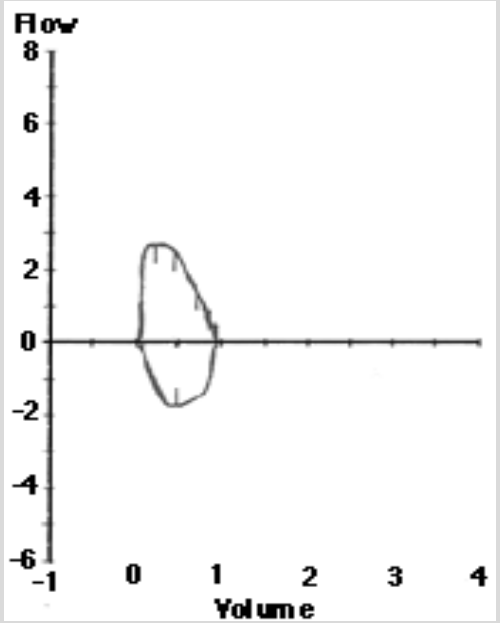
- This pattern is seen with:
  - Asthma
  - Chronic obstructive pulmonary disease (COPD)
    - Chronic bronchitis
    - Emphysema

The difference with asthma vs. COPD is that asthma is (typically) **REVERSIBLE**

Diagnosing Asthma: Testing Expiratory Airflow

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- **Restrictive Flow Volume Loop**



- **Pattern looks normal with a shift toward lower lung volumes and flow rates**
- *A reduced FVC with a normal FEV1/FVC is suggestive of a restrictive disorder*
  - Pulmonary fibrosis
  - *Pulmonary edema*
  - Interstitial pneumonia
  - *Diaphragm paralysis*



Diagnosing Asthma: Testing Expiratory Airflow

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- **Bronchoprovocation Challenge**
  - If a patient has the history suggestive of asthma and presents asymptomatic, this test can be implemented
    - Baseline FEV_1 is measured
    - Methacholine is inhaled to induce an acute exacerbation
      - Cholinomimetic agent; stimulates muscarinic receptors
    - FEV_1 is re-tested
      - **Positive test:** a decrease of 20% or more from baseline
    - The patient is treated with a bronchodilator
    - The patient is retested to show return of FEV1 to baseline



Diagnosing Asthma: Other

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- **Flow-Volume Loops**
  - Aids in detection of upper-airway obstruction
- **Lung Volume Measurements**
  - Usually normal
  - Air trapping seen during exacerbation
  - Vital capacity can be reduced
- **Airway Resistance**
  - Determined by plethysmography (body-box)
  - More sensitive measurement than FEV_1

Diagnosing Asthma: Other

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- **Diffusion Capacity**
  - Usually normal, may be elevated
    - Elevation indicated recruitment of capillaries from higher pulmonary arterial pressure
- **Hypoxemia** *RARE*
  - Usually indicated severe obstruction when present
  - Alveolar-arterial oxygen difference often elevated
- **Hypocapnia and respiratory alkalosis**
  - More common finding on arterial blood gas (ABG)
  - Normal (or increasing) pCO_2 during severe exacerbation indicated pending respiratory failure

Diagnosing Asthma: Excluding Differentials

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- **Infants and Children**
  - **Upper airway disease** (Allergic Rhinitis and Sinusitis)
  - **Obstruction involving large airways** (Foreign body, vocal cord dysfunction, vascular rings or laryngeal webs, Laryngotracheomalacia, tracheal stenosis, bronchostenosis, **Enlarged lymph nodes or tumor**
  - **Obstruction involving large airways** (Viral bronchiolitis or obliterative bronchiolitis, Cystic fibrosis, bronchopulmonary dysplasia, congenital heart disease)
  - **Other** (recurrent cough not due to asthma, aspiration from swallowing mechanism dysfunction or GERD)

Diagnosing Asthma: Excluding Differentials

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- **Adults**
  - **Chronic Obstructive Pulmonary Disease** (COPD)
  - **Congestive Heart Failure** (CHF)
  - **Pulmonary Embolism** (PE)
  - **Laryngeal dysfunction**
  - **Mechanical obstruction of the airways** (tumors)
  - **Pulmonary infiltrates with eosinophils**
  - **Cough secondary to drugs** (ACE inhibitors)
  - **Vocal cord dysfunction**

Precipitating Factors for Asthma Exacerbations

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- **Environmental Allergies**
  - Pollen, trees, gases, molds
- **Sleep**
  - Nocturnal symptoms common (3-4 AM)
    - Often overlooked because occurs so often
  - Also, early morning awakening
  - Endogenous circadian rhythms underlie nocturnal reduction in airway caliber
    - Decreased levels of catecholamines
    - Increased level of histamine

Precipitating Factors for Asthma Exacerbations

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- **Gastroesophageal reflux**
  - May exacerbate asthma symptoms
  - Vagally mediated reflexes involved
- **Sinus disease**
  - Persistent posterior drainage of nasal mucus
  - Sinusitis often found in patients with asthma

Treatment → PREVENTION

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- There is no cure for asthma
- The primary goal is **control of symptoms** through **reduction exacerbations** by limiting exposure of precipitating factors and medication
- When symptoms are not in control, a conversation must be had about exposures and use of medications.
  - Poor medication compliance is a major factor in uncontrolled asthmatics

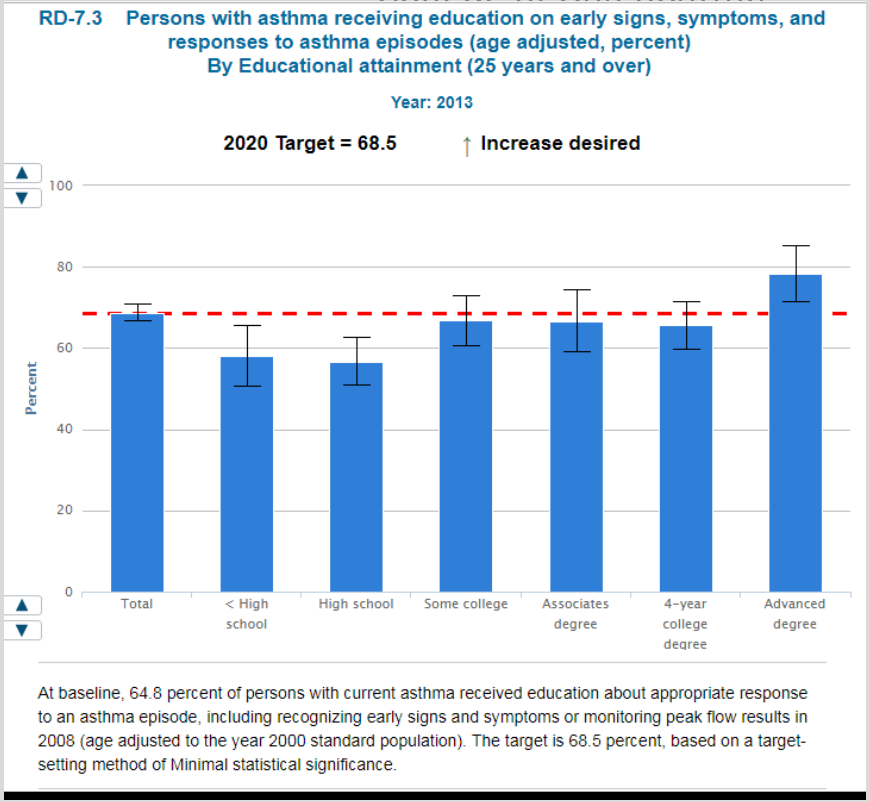


Treatment → Education

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- The United State Task Force published a review article looking at a variety different levels of prevention measures (23 studies)
  - **Mild**: mattress and pillow covers
  - **Moderate**: teaching how to use bedding covers, household cleaning supplies, dehumidifiers, pest management and minor home structural repairs
  - **Major**: structural (remove carpet, replace ventilation, etc.)
- Economic Review: "The combination of minor to moderate environmental remediation with an educational component provides good value for the money invested based on improvement in symptom free days & savings from averted costs of asthma care and improvement in productivity."



Checklist for uncontrolled asthma

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- A – allergy and adherence to therapy
- I – inflammation and infection
- R – rhinitis and rhinosinusitis
- E – exercise and error in diagnosis
- S – smoking
- M – medications
- O – occupational, hobby, and school
- G – gastroesophageal reflux

Courtesy of G. Volcheck, K. Lim, Mayo Clinic

STATUS ASTHMATICUS

- An acute asthma exacerbation that does not respond to appropriate conventional treatment
- Implies a severe life-threating episode
  - Mortality is higher for minorities
- **Treatment**
  - IV corticosteroids
  - Inhaled β-2 agonists
  - ? IV $MgSO_4$
  - Heliox
  - Noninvasive ventilation (CPAP/BiPAP)
  - Mechanical ventilation



Summary Slide

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- Asthma is a chronic heterogeneous condition of the airway that presents with episodes of wheezing, breathlessness, chest tightness, nighttime or early morning cough
- Asthma has a significant impact on activities of daily living
  - Effects more African American people than other races
  - Disproportionally effects people below the poverty line and people living in inner cities
  - Boys more than girl; women more than men
- Asthma is classified as allergic or idiosyncratic (non-allergic)
  - Allergic is associated with higher levels of IgE and atopy
  - IgE stimulates mast cells to release histamine, cytokines and leukotrienes leading to the symptoms of asthma
- The environment and genetics play a key role in the development of asthma
- Inflammation + hyperresonspiveness → airway obstruction → asthma symptoms

Summary Slide

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- **Diagnosis**:
  1. Detailed medical history and physical that is suggestive of asthma
  2. Demonstration of variable expiratory airflow limitation
     - "the scoop", ≥ 20% decrease with methacholine challenge, reversible!
  3. Exclusion of alternative diagnoses
- Status asthmatics is defined by an acute asthma exacerbation that FAILS to respond to appropriate treatment
  - Red flags: "silent chest", rise in pCO_2, drowsiness, cyanosis, and failure to improve with treatment
- The goal of asthma treatment is CONTROL by prevention of acute exacerbations
  - Avoiding irritants, medication compliance
  - Education is key to preventing acute asthma attacks